

# Cloud Data Platform

## Technical Datasheet

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AWS + Databricks Lakehouse  
End-to-End Real Data Pipeline

*Version 1.1*

*Phase: P0 - Decisions and Project Bootstrap*

*Status: DECIDED / SELECTED / PENDING MANUAL VERIFICATION*

## A. PROJECT OVERVIEW

The Cloud Data Platform is an end-to-end, portfolio-scale data engineering project that demonstrates production-grade practices for ingesting real external data, transforming it through a medallion architecture, applying data quality controls, and exposing business-oriented analytics.

The platform runs on AWS and Databricks. It ingests from two public REST APIs (World Bank Indicators, FRED Economic Series) and one operational database (Amazon RDS PostgreSQL via Fivetran). Data flows through immutable S3 raw storage, Bronze/Raw Fivetran layers, dbt-managed Silver and Gold transformations, and is governed by Unity Catalog. Orchestration uses Amazon MWAA (Airflow) as an ephemeral environment created and destroyed per work window. Infrastructure is codified in Terraform with GitHub Actions CI/CD using OIDC authentication.

## B. ARCHITECTURE

### Architecture Diagram

See docs/architecture1.png for the visual architecture reference. The diagram shows two distinct ingestion paths converging at dbt transformations:

Path A - API Ingestion:

- World Bank API -> Python Client -> MWAA/Airflow -> S3 Immutable Raw -> Databricks Bronze
- FRED API -> Python Client -> MWAA/Airflow -> S3 Immutable Raw -> Databricks Bronze

Path B - Database Ingestion:

- Amazon RDS PostgreSQL (macro\_watchlist\_db) -> Fivetran (PostgreSQL connector) -> Databricks (schema: raw\_fivetran, prefix to be confirmed in P8)

Shared Transformation Layer:

- Bronze + raw\_fivetran (prefix TBC in P8) -> dbt Silver (staging) -> dbt Gold (marts) -> Databricks AI/BI

Quarantine (Hybrid per ADR-002):

- Structural failures (malformed JSON, schema mismatch) -> S3 quarantine prefix
- Semantic failures (business rules, PK/FK, domain values) -> Unity Catalog quarantine schema

## C. DATA SOURCES

### World Bank Indicators (API 1)

| Property           | Value                                                    |
|--------------------|----------------------------------------------------------|
| Source Name        | world_bank_indicators                                    |
| Type               | Public REST API                                          |
| Provider           | World Bank                                               |
| Base URL           | https://api.worldbank.org/v2                             |
| Authentication     | None                                                     |
| Pagination         | page + per_page (max 500)                                |
| Response Format    | JSON                                                     |
| Key Endpoints      | /country/{country}/indicator/{indicator}                 |
| Initial Indicators | NY.GDP.MKTP.KD.ZG, FP.CPI.TOTL.ZG, SL.UEM.TOTL.ZS        |
| Expected Volume    | ~6,000 observations/run (3 ind x 200 countries x 10 yrs) |

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| Data Grain         | One observation per country + indicator + year                  |
| Raw S3 Path        | s3://<bucket>/api/world_bank_indicators/ingest_date=YYYY-MM-DD/ |
| Bronze Destination | <catalog>.bronze.world_bank_indicators                          |
| Status             | Selected                                                        |

## FRED Economic Series (API 2)

| Property           | Value                                                          |
|--------------------|----------------------------------------------------------------|
| Source Name        | fred_economic_series                                           |
| Type               | Public REST API                                                |
| Provider           | Federal Reserve Bank of St. Louis                              |
| Base URL           | https://api.stlouisfed.org/fred                                |
| Authentication     | API Key (via Secrets Manager)                                  |
| Pagination         | limit + offset                                                 |
| Response Format    | JSON                                                           |
| Key Endpoints      | /series/observations                                           |
| Initial Series     | CPIAUCSL, UNRATE, FEDFUNDS                                     |
| Expected Volume    | ~780 observations/run (3 series x 5 yrs monthly)               |
| Data Grain         | One observation per series + observation date                  |
| Raw S3 Path        | s3://<bucket>/api/fred_economic_series/ingest_date=YYYY-MM-DD/ |
| Bronze Destination | <catalog>.bronze.fred_economic_series                          |
| Status             | Selected                                                       |

## Amazon RDS PostgreSQL via Fivetran (Operational DB)

| Property             | Value                                                                |
|----------------------|----------------------------------------------------------------------|
| Source Name          | macro_watchlist_db                                                   |
| Type                 | Operational PostgreSQL Database                                      |
| Provider             | Amazon Web Services                                                  |
| Database             | Amazon RDS for PostgreSQL                                            |
| Fivetran Connector   | PostgreSQL                                                           |
| Authentication       | DB Credentials (via Secrets Manager)                                 |
| Sync Frequency       | Daily / On-demand via Airflow                                        |
| Expected Volume      | < 1,000 rows total                                                   |
| Data Grain           | One row per entity (country, indicator, alert rule)                  |
| Destination          | dwh_dev.raw_fivetran.<table> (Fivetran prefix to be confirmed in P8) |
| Incremental Strategy | Query-Based using xmin                                               |
| Tables               | watchlist_country, watchlist_indicator, alert_rule                   |
| Status               | Selected                                                             |

## D. INITIAL DATA CONTRACTS

### World Bank Indicators

| Indicator Code    | Name                                   | Unit    | Frequency | Historical Range |
|-------------------|----------------------------------------|---------|-----------|------------------|
| NY.GDP.MKTP.KD.ZG | GDP Growth (annual %)                  | Percent | Annual    | 10 years         |
| FP.CPI.TOTL.ZG    | Inflation, consumer prices (annual %)  | Percent | Annual    | 10 years         |
| SL.UEM.TOTL.ZS    | Unemployment, total (% of labor force) | Percent | Annual    | 10 years         |

### FRED Economic Series

| Series ID | Name                                         | Unit              | Frequency | Historical Range |
|-----------|----------------------------------------------|-------------------|-----------|------------------|
| CPIAUCSL  | Consumer Price Index for All Urban Consumers | Index 1982-84=100 | Monthly   | 5 years          |
| UNRATE    | Civilian Unemployment Rate                   | Percent           | Monthly   | 5 years          |
| FEDFUNDS  | Federal Funds Effective Rate                 | Percent           | Monthly   | 5 years          |

### RDS PostgreSQL Tables (macro\_watchlist\_db)

| Table               | Key Fields                                                         | Purpose                                |
|---------------------|--------------------------------------------------------------------|----------------------------------------|
| watchlist_country   | country_id, country_code, country_name, region, active, added      | Countries to monitor                   |
| watchlist_indicator | indicator_id, source_system, source_key, indicator_name, unit      | Indicators to track (WB/FRED)          |
| alert_rule          | alert_rule_id, country_code, indicator_id, threshold, direction, a | Alert thresholds per country/indicator |

## E. DATA LAYERS

The platform implements a medallion architecture with additional quarantine and snapshot layers:

- S3 Raw/Landing: Immutable source payloads. API ingestions create new objects per ingestion date; never overwrite. Partitioned by ingest\_date=YYYY-MM-DD.
- Bronze: Data close to source plus technical metadata: \_batch\_id, \_ingested\_at, \_source\_file, \_source\_system, \_batch\_date. Loaded via COPY INTO (ADR-001).
- raw\_fivetran: Fivetran-managed landing area for operational database. Direct write to Databricks, no S3 intermediary (D-002). Schema prefix to be confirmed in P8.
- Silver: Parsing, types, deduplication, normalization, business rules. Implemented as dbt staging models.
- Gold: Analytical dimensions and facts in dbt marts. Stable business definitions for dashboards.
- Quarantine: Rejected records with diagnostic metadata. Structural -> S3 quarantine; Semantic -> Unity Catalog quarantine schema (ADR-002).
- Snapshots: dbt snapshot history using SCD2 where historical tracking is required (e.g., watchlist changes).

## F. UNITY CATALOG

Governance through Unity Catalog with the following structure:

| Catalog | Schemas | Purpose                     |
|---------|---------|-----------------------------|
| dwh_dev | bronze  | Raw Bronze tables from APIs |
|         | silver  | Staging/cleaned models      |

|  |              |                                                                           |
|--|--------------|---------------------------------------------------------------------------|
|  | gold         | Analytical marts                                                          |
|  | quarantine   | Semantic rejection tables                                                 |
|  | snapshots    | SCD2 history tables                                                       |
|  | raw_fivetran | Fivetran-landed tables (macro_watchlist_db); prefix to be confirmed in P8 |

Permissions follow least privilege: pipeline identities write where needed; BI consumers read; developers get minimal development access. S3 access via Unity Catalog storage credentials and external locations.

## G. ORCHESTRATION

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### MWAA / Airflow

Amazon MWAA runs Apache Airflow as an ephemeral environment (ADR-004, D-005, D-006):

- Created/destroyed via manual GitHub Actions workflows (not on merge to main)
- Private subnets with one NAT Gateway for portfolio cost/realism balance
- DAGs and requirements.txt live in persistent S3 (persistent stack)
- Connections/variables from AWS Secrets Manager (Airflow secrets backend)
- Airflow metadata DB is ephemeral; persistent facts in S3/Databricks
- Pre-flight DAG import/parse test runs before provisioning

### Daily Pipeline Flow (Conceptual)

The intended DAG flow (to be implemented in P7):

- 1. API ingestion (World Bank, FRED) -> write raw JSON to S3
- 2. Check S3 raw availability
- 3. Trigger / wait for Fivetran sync (when enabled)
- 4. Validate raw/landing availability
- 5. Load Bronze via COPY INTO (ADR-001)
- 6. dbt build (runner: MWAA -> SQL Warehouse, ADR-003)
- 7. dbt test / data quality checks
- 8. Confirm marts available

## H. DBT

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### Layers

- Silver: staging models for each source (world\_bank, fred, macro\_watchlist\_db)
- Gold: dimensional models (dim\_country, dim\_indicator, dim\_fred\_series, dim\_alert\_rule) and fact models (fact\_macro\_indicator, fact\_fred\_observation, fact\_alert\_evaluation)
- Tests: not-null, unique, relationships, accepted-values, freshness/recency, custom business rules
- Snapshots: SCD2 for entities with meaningful history (e.g., watchlist\_indicator.active changes)

### Execution

dbt runs from MWAA execution context against Databricks SQL Warehouse (ADR-003). Connection via environment variables; no committed profiles.yml with real values. dbt-databricks adapter with SQL Warehouse HTTP path.

## I. FIVETRAN

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Managed ingestion path for the operational database:

- Source: Amazon RDS PostgreSQL (macro\_watchlist\_db)
- Connector: PostgreSQL
- Destination: Databricks (Unity Catalog) -> raw\_fivetran schema (prefix to be confirmed in P8)
- Incremental: Query-Based using xmin
- Security: TLS, dedicated read-only Fivetran user, restricted network access, no 0.0.0.0/0 exposure

- Optional per D-001: core platform functions without Fivetran; P8 can be formally skipped if plan lacks Databricks destination

## J. QUARANTINE (ADR-002)

Hybrid quarantine boundary accepted:

| Layer                      | Validation Type     | Failures Diverted To           | Examples                                                           |
|----------------------------|---------------------|--------------------------------|--------------------------------------------------------------------|
| Ingestion (Python/Airflow) | Structural / Schema | S3 quarantine prefix           | Malformed JSON, missing top-level fields, schema mismatch          |
| Transformation (dbt)       | Semantic / Business | Unity Catalog quarantine scher | Null PK, FK violation, domain values, business rules, completeness |

All rejected records retain: source\_system, source\_file, batch\_id, rejection\_timestamp, rejection\_reason. No silent drops.

## K. AWS SERVICES (Approved Scope)

Only the following AWS services are in scope. Others (Redshift, Glue, EMR, Athena, Kinesis, Lambda, DynamoDB, ECR, ECS) are explicitly excluded.

- S3: Raw/landing bucket (versioned, SSE-S3, prevent\_destroy), DAG bucket
- MWAA: Ephemeral Airflow environment (private subnets, NAT Gateway)
- VPC/Networking: Private subnets, NAT Gateway (single AZ for portfolio), Security Groups
- Secrets Manager: API keys, DB credentials, Fivetran credentials, Databricks tokens
- IAM: Least-privilege roles for MWAA, Databricks, Fivetran, CI/CD
- RDS PostgreSQL: Operational source for macro\_watchlist\_db (ADR-005 exception)
- AWS Budgets: Monthly cost guardrail (\$10/month) with alerts
- CloudWatch Logs: MWAA and infrastructure logging

## L. DATABRICKS

- Unity Catalog: Governance layer: catalog dwh\_dev, schemas bronze/silver/gold/quarantine/snapshots/raw\_fivetran
- SQL Warehouse: Smallest size, short auto-stop. Used for COPY INTO (Bronze) and dbt execution (Silver/Gold)
- Bronze: Delta tables loaded via COPY INTO from S3 raw
- raw\_fivetran: Fivetran direct write destination (schema prefix TBC in P8)
- Silver: dbt staging models
- Gold: dbt mart models
- Quarantine: Semantic rejection Delta tables

## M. ARCHITECTURAL DECISIONS (ADRs)

| ADR     | Topic                | Decision                                                          | Status   |
|---------|----------------------|-------------------------------------------------------------------|----------|
| ADR-001 | Bronze Loading       | COPY INTO via Databricks SQL Warehouse                            | Accepted |
| ADR-002 | Quarantine Boundary  | Hybrid: ingestion structural -> S3; dbt semantic -> UC quarantine | Accepted |
| ADR-003 | dbt Runner           | dbt from MWAA execution context against SQL Warehouse             | Accepted |
| ADR-004 | MWAA Operating Model | GitHub Actions create/destroy, private subnets + NAT              | Accepted |
| ADR-005 | RDS Exception        | Amazon RDS PostgreSQL only as operational source for Fivetran     | Accepted |

## N. BUSINESS QUESTIONS

### BQ-001: Macro Watchlist Evolution

How have GDP growth, inflation, and unemployment evolved over time for countries on the macro watchlist?

Grain: country + indicator + year | Dimensions: country, region, indicator, year | Metrics: value, latest, absolute/percent change, trend | Range: 10 years | Sources: World Bank + macro\_watchlist\_db | Status: Approved

### BQ-002: US Macroeconomic Trends

What are the latest trends in the selected US macroeconomic series, and how do their values change over time?

Grain: series + observation date | Dimensions: series, date, frequency, unit | Metrics: latest, previous, absolute/percent



change, trend | Range: 5 years | Source: FRED | Status: Approved

### **BQ-003: Alert Rule Breaches**

Which configured macroeconomic alert rules are currently breached, and for which countries or indicators?

Grain: country + indicator + alert rule + latest observation | Dimensions: country, region, indicator, source\_system, direction, active | Metrics: latest value, threshold, difference, breach status, latest date | Range: latest | Sources: World Bank + macro\_watchlist\_db | Status: Approved

### **Data Quality Questions (DQ-001 through DQ-007)**

Pipeline status, last successful batch timestamp, rows ingested per source/batch, rows quarantined per source/batch, failing dbt tests, data freshness vs SLA, source API health (response time, error rate, last success).

## O. SECURITY

- No secrets in Git. All credentials via AWS Secrets Manager, GitHub OIDC, or runtime secret mechanisms.
- Secret management: FRED API key, RDS credentials, Fivetran credentials, Databricks auth via approved mechanisms.
- Least privilege IAM: pipeline identities write only where necessary; BI consumers read-only; developers minimal.
- Fivetran: read-only database user, TLS, restricted network access (Fivetran IPs only).
- RDS: no public access (0.0.0.0/0), security group restricted to Fivetran CIDR ranges.
- AWS root account: MFA enabled. Administrative identity is non-root.
- GitHub Actions: OIDC authentication to AWS. No long-lived AWS access keys.
- Unity Catalog: storage credentials and external locations for S3 access (no ad-hoc credentials in code).

## P. COST CONTROL

### AWS Budget Guardrail

| Parameter           | Value                                                   |
|---------------------|---------------------------------------------------------|
| Budget Name         | monthly-cost-guardrail                                  |
| Amount              | 10 USD/month                                            |
| Alerts (Actual)     | 50%, 80%, 100%                                          |
| Alerts (Forecasted) | 100%                                                    |
| Type                | Alerting/monitoring guardrail (NOT a hard spending cap) |

Note: This is an alerting guardrail. It does not enforce a hard stop. Afonso must create the actual AWS Budget manually before any billable apply (manual prerequisite).

### Cost Awareness

- MWAA: billed while environment exists (~\$0.49/hr for small); cannot be paused, only destroyed. Creation ~20-30 min, destruction ~15-20 min (both billed).
- NAT Gateway: ~\$45/month per AZ (single AZ for portfolio). Primary cost driver alongside MWAA.
- Databricks SQL Warehouse: smallest size, short auto-stop (e.g., 10 min). Never leave running between sessions.
- Ephemeral MWAA model: environments created only for planned work windows; destroyed immediately after evidence collection.
- Post-destroy verification: confirm nothing billable remains via Console/Cost Explorer.

## Q. CURRENT PHASE

### P0 - Decisions and Project Bootstrap

All architectural decisions (ADR-001 through ADR-005) are Accepted. All data sources (World Bank, FRED, RDS PostgreSQL) are Selected. Business questions (BQ-001, BQ-002, BQ-003) are Approved. P1 implementation has NOT started.

## R. STATUS SUMMARY

### DECIDED (Architecture & Source Choices)

- Dual ingestion paths: API (S3 -> Bronze) and Database (Fivetran -> raw\_fivetran, prefix TBC in P8)

- Bronze loading: COPY INTO via SQL Warehouse (ADR-001)
- Quarantine: Hybrid boundary (ADR-002)
- dbt runner: MWAA -> SQL Warehouse (ADR-003)
- MWAA: Ephemeral, GitHub Actions create/destroy, private + NAT (ADR-004)
- RDS PostgreSQL: Approved only as operational source for Fivetran (ADR-005)

### SELECTED (Sources)

- World Bank Indicators API (no auth, page/per\_page pagination)
- FRED Economic Series API (API key, limit/offset pagination)
- Amazon RDS PostgreSQL (macro\_watchlist\_db) via Fivetran PostgreSQL connector

### PENDING MANUAL VERIFICATION

- Fivetran plan supports Databricks destination
- Databricks workspace supports storage credentials + external locations on S3 + SQL Warehouse
- AWS root MFA, non-root admin identity, AWS Budget with alerts created

### NOT YET IMPLEMENTED

- RDS PostgreSQL instance
- Fivetran connector configuration
- MWAA environment
- S3 buckets
- Databricks catalogs/schemas/tables
- Terraform infrastructure
- dbt project and models
- Airflow DAGs
- GitHub Actions workflows